

INDRAJITH MP

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TECHNICAL SKILLS

AI/ML: Python, PyTorch, TensorFlow, Keras, Scikit-learn, Deep Learning, CNN, Reinforcement Learning, DQN, Computer Vision, OpenCV, MediaPipe, NLP, LLM, Transfer Learning, Feature Engineering, Model Optimization
Robotics & Embedded: ROS2, Raspberry Pi Pico, Arduino, Microcontroller Programming, Sensor Integration (Ultrasonic, IMU, Encoders), Autonomous Navigation, Control Systems, Sensor Fusion, GPIO, PWM, I2C, UART
Development & Deployment: C/C++, TypeScript, JavaScript, Docker, Kubernetes, CI/CD, GitHub Actions, Git, Linux, AWS, Cloudflare Workers, Flask, REST APIs, NumPy, Pandas, Matplotlib

EXPERIENCE

Moonpreneur (Silicon Valley-based Startup)

Feb 2025 – Present

Robotics Project Guide Intern

Remote

- Built and deployed **sensor-based robotic systems** using **Raspberry Pi Pico**, integrating **ultrasonic sensors, IMU, and encoders** via GPIO, PWM, and I2C for real-time navigation, actuator control, and AI-driven decision logic.
- Delivered **15+ technical sessions** to **50+ students** covering robotics, embedded systems, and **AI/ML fundamentals** including supervised learning, neural networks, and computer vision.
- Developed and taught **data science curriculum** with hands-on Python, Pandas, NumPy, and scikit-learn modules; guided students through end-to-end ML pipelines from data preprocessing to model deployment.
- Created **computer vision coursework** using **OpenCV and MediaPipe** for real-time object detection and hand tracking.

AI/ML Summer Intern

June 2024 – July 2024

Allsoft Solutions (IBM Business Partner)

Remote

- Engineered end-to-end **Fake News Detection system** using **NLP** and supervised ML, preprocessing **50,000+ data points** from LIAR dataset and Indian news sources with **99.2% data cleaning accuracy**
- Trained ensemble of classifiers (Logistic Regression, SVM, XGBoost, Naive Bayes) using GridSearchCV hyperparameter tuning, achieving **97% test accuracy** and **0.96 F1-score** on validation data

PROJECTS

Failure-Aware Vision – ML Perception Engine | *FastAPI, ROS, OpenCV, WebSockets*

February 2026

- Engineered real-time frame-level anomaly scoring (blur, brightness, entropy, freeze detection) at **30Hz** using OpenCV, modeling reliability as a continuous confidence score with **asymmetric temporal decay**
- Designed **trust-based safety state machine** with bounded ML influence and deterministic policy gating, achieving **zero false-safe resumptions** across **5 corruption types** in simulation validation
- Refactored ROS prototype into **FastAPI + WebSocket** deployment architecture with live telemetry streaming, cloud-hosted on Render with interactive corruption injection and CSV diagnostics export

Intentra – AI Prompt Optimization Platform | *TypeScript, LLM Orchestration, Cloudflare*

January 2026

- Built multi-stage AI prompt engineering platform transforming ambiguous inputs into structured prompts, improving LLM output reliability by **67%** with quality scoring algorithm
- Implemented **LLM orchestration** with automatic failover between Gemini 1.5 and Groq Llama 3.1, achieving **99.9% uptime** through quota-aware load balancing with **less than 100ms latency**
- Architected serverless backend using **Cloudflare Workers** with session tracking and rate limiting

Gesture-Controlled Human-Machine Interface | *Python, OpenCV, MediaPipe*

September 2025

- Engineered real-time hand gesture recognition using **MediaPipe** achieving **95% accuracy at 30 FPS** for touchless control of media, presentations, and cursor navigation across **8 gesture classes**
- Integrated **Hugging Face Stable Diffusion API** to convert hand-drawn sketches into digital artwork with **less than 2 second inference latency**, processing **100+ image generations**
- Designed modular gesture mapping with Kalman filtering, reducing false positives by **34%** for deployment on robotic platforms

EDUCATION

Bachelor of Technology in Computer Science Engineering

Expected May 2026

Lovely Professional University – CGPA: 9.08/10.0

Punjab, India

- Relevant Coursework:** Machine Learning, Deep Learning, Computer Vision, Data Structures & Algorithms, Operating Systems, Computer Networks

CERTIFICATIONS

Generative AI with Large Language Models (DeepLearning.AI) | Algorithms Specialization (Stanford University)